Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Constructing Parallel Lines with a Set Square and Ruler

Two straight lines are parallel when they always stay the same distance apart, so they never meet no matter how far you extend them. The two rails of a railway track are parallel, and so are the two upright sides of a ladder. You already know how to draw a straight line with a ruler. Now you will learn how to construct a line that is parallel to a line you have already drawn, using two simple tools: a set square and a ruler.

The trick is to let the set square slide along the ruler, keeping the ruler fixed. Because the set square is a right-angled triangle, when you slide it its edge never changes direction — so the line you draw is parallel to the one you started with.

To construct a line through a point P that is parallel to a given line AB:

1. Draw the given line and label its ends A and B. Mark clearly the point P through which the new parallel line must pass.
2. Place the set square so that one of its straight edges lies exactly on top of the line AB.
3. Hold the ruler firmly against the side of the set square, so that the ruler does not move.
4. Keeping the ruler pressed against the table, slide the set square along the ruler until its edge reaches point P.
5. Now hold the set square firmly and draw a straight line along its edge through P.

The line you have drawn through P is parallel to AB.

Construct a line through point P that is parallel to the given line AB.

Step 1 — Draw a line about 8 cm long and label its ends A and B. Mark a point P about 3 cm above the middle of the line.

Step 2 — Place the set square so one of its straight edges lies exactly on line AB. The edge must touch the line along its whole length.

Step 3 — Hold the ruler firmly against the side of the set square, keeping it fixed in position.

Step 4 — Slide the set square up the ruler until its edge passes through point P.

Step 5 — Press the set square firmly and draw a straight line through P along its edge. Label this new line.

Check your work: measure the gap between AB and your new line at two different places with your ruler. The two gaps should be equal. That is the sign that the two lines are parallel — they stay the same distance apart, so they will never meet.

Practice

1. Draw a straight line AB about 8 cm long. Mark a point P about 3 cm above the line. Using only a set square and a ruler, construct the line through P that is parallel to AB. Write down, in order, the steps you took.
2. Draw a line CD about 6 cm long. Mark a point Q below the line. Construct the line through Q that is parallel to CD using a set square and ruler. Now measure the gap between the two lines at two different places. What should you find, and what does this tell you?
3. Draw a line AB about 7 cm long. Mark a point P above the line and a point Q below the line. Construct a line through P parallel to AB, and a separate line through Q parallel to AB. What do you notice about the three lines, and why?

Section 2: Drawing Perpendicular Lines

Two lines are perpendicular when they meet at a right angle — that is, at exactly 90°. You have seen this idea when drawing squares, rectangles and right angles. With your mathematical set you can draw

a perpendicular to any straight line, and you do it with just a ruler and a set square.

Here is the key idea. Your set square has one corner that is a perfect right angle, and the two edges that meet at that corner are perpendicular to each other. If you slide one of those edges along a straight line, the other edge always stays at 90° to it. So, wherever you stop, the edge that is not lying on the line is a perpendicular line — you just draw along it.

To draw a perpendicular to a given line:

Place the ruler so one of its long edges lies exactly along your given line. Hold it still.
Place the set square so that one of its shorter edges lies flat against the ruler, with the right-angle corner pressed against the ruler's edge.
Draw along the other short edge of the set square. This edge is at right angles to the line, so the line you draw is perpendicular to it.

Mark the point where the two lines meet and label the right angle so you can see it is 90°.

Example. Draw a line AB, and then construct a perpendicular to it.

Step 1. Draw the line AB with your ruler and label its ends A and B.
Step 2. Place your ruler so one long edge lies exactly along AB. Hold the ruler firmly so it does not slip.
Step 3. Place your set square against the ruler, with its right-angle corner pressed to the ruler's edge and one of its short edges lying flat along the ruler.
Step 4. Slide the set square along the ruler until the edge you will draw from is where you want the perpendicular to stand, then press it down.
Step 5. Draw a line along the other short edge of the set square. Extend it with your ruler so it is clearly a straight line crossing AB.
Step 6. Mark the point where the two lines meet, and draw a small square in the corner to show the right angle.

The line you have drawn meets AB at a right angle, so it is perpendicular to AB. Check your work by looking at the corner: the two lines form a clean 90° angle.

Practice

1. Draw a horizontal line PQ on your page. Using your ruler and set square, construct a line perpendicular to PQ that passes through a point you choose on PQ. Name the point where the two lines meet.
2. Draw a line segment AB. Construct a perpendicular to AB at its midpoint — that is, halfway between A and B. Measure the four angles where your new line crosses AB. What do you notice about them?

Section 3: Cutting a Line Exactly in Half

When you draw a line, you can find its midpoint by measuring with a ruler. But in construction we find that point using only your mathematical set — a pair of compasses and a straight edge. This is called **bisecting** the line segment: cutting it into two equal halves.

To bisect a line segment you draw a line that crosses it at its exact middle and meets it at a right angle. That line is called the **perpendicular bisector** of the segment. The point where it crosses the segment is the midpoint, and it splits the original segment into two equal parts.

Here is the idea. A point that is the same distance from the two ends of a segment must lie on the line that cuts the segment exactly in half. So we find two such points — one above the segment and one below it — and join them. The line through those two points is the perpendicular bisector, and where it crosses the segment is the midpoint.

Let us bisect a line segment AB.

1. Draw the segment AB with your ruler. Leave a little space above and below it, because the compass work happens there.
2. Put the point of your compasses on A. Open them so the gap is more than half the length of AB — clearly more than half, not just a little. Draw an arc that passes above and below the segment, near where the middle will be.
3. Without changing the compass opening, move the point to B. Draw a second arc. The two arcs meet at two points: one above the segment and one below it.
4. With your ruler, draw a straight line through the two points where the arcs crossed. This line crosses AB at its midpoint. Label that point M.

The segment is now cut into two equal halves: AM and MB are equal in length, and the line you drew meets AB at a right angle.

Practice

1. Draw a line segment CD that is 8 cm long. Bisect it with your compasses and ruler. Mark the midpoint M. What are the lengths of CM and MD?
2. When you bisect a line segment, why must the compass opening be more than half the length of the segment (not exactly half, and not less than half)?
3. You bisect a line segment and label the midpoint M. State two things that are always true about the line you drew that crosses the segment at M.

Section 4: Constructing a Right Angle (90°)

A right angle is the angle that measures exactly 90°. You meet it at the corner of a page, a doorframe or a set square. You can build one yourself using only a ruler and a pair of compasses.

The key idea is this: if a line stands straight up on another line, the two angles it makes at the meeting point are both right angles. We say the standing line is *perpendicular* to the first line, and the angle where they meet is 90°. So constructing a 90° angle is the same as constructing a perpendicular to a line at a chosen point on it.

Here is the plan. You start with a line and mark a point where the right angle should sit. Then you draw a sequence of compass arcs — some cutting the line, some rising above it — and the last point where two arcs meet gives you the direction of the perpendicular. A final line drawn from that point to your marked point completes the right angle.

Worked example — construct a 90° angle at point O on a line.

Use a sharp pencil, a ruler and a pair of compasses. Keep the radius of your compasses steady while you use it.

Step 1. Draw a straight line across your page. Mark a point O on it. This is where the right angle will sit.

Step 2. Place the compass point on O. Open the compasses to any convenient radius — about half the width of your page works well. Draw a complete circle with centre O. Where that circle crosses your straight line, you get two points; label them A (on the left) and B (on the right). The distances OA and OB are equal because they are both radii of the same circle.

Step 3. Put the compass point on A. Without changing the radius, draw a large arc above the line, roughly in the middle. Make it long enough that a second arc can cross it.

Step 4. Now put the compass point on B. Still without changing the radius, draw another arc above the line so that it crosses the arc from Step 3. Label the point where the two arcs cross C.

Step 5. Use your ruler to draw a straight line from C to O. This line is perpendicular to your original line, so it stands straight up on it.

Step 6. Check your work: the angle between the original line and the line OC — on either side of O — should look like the corner of a page. Measure it with a protractor if you have one; it should read 90°.

Why this works: the compasses were kept at the same radius throughout, so OA = OB = AC = BC. With all four sides equal, figure AOBC is a rhombus, and the line OC is the perpendicular bisector that splits the straight angle AOB of 180° exactly in half, giving 90° on each side.

Practice

1. On a fresh line, mark a point O and construct a 90° angle at O using your ruler and compasses only. Then check your angle by measurement.
2. In the construction of a 90° angle, the compasses are kept at the same radius in every step. Explain in one or two sentences why this matters.

Section 5: Drawing an Angle of 60°

An angle of 60° is one you will meet again and again, because it sits inside equilateral triangles and regular hexagons. The good news is that you can draw it with just your compass and ruler — no protractor needed. The secret is the equilateral triangle hiding inside the construction: when you make three arcs of the same radius that join end to end, the angle they enclose is exactly 60°.

Construct an angle of 60° at the point A on a line.

Step 1. Draw a straight line and mark a point A on it. This is where the 60° angle will sit.

Step 2. Open your compass to any convenient radius. Put the point of the compass on A and draw an arc that crosses the line. Call the point where the arc meets the line B. The radius AB is your working radius.

Step 3. Without changing the radius, put the compass point on B and draw a second arc, cutting the first arc. Call the point where the two arcs cross C.

Step 4. Use your ruler to join A to C with a straight line.

The angle BAC is 60°. Why? Because AB, BC and AC are all equal — each is the same compass radius — so triangle ABC is equilateral, and every angle of an equilateral triangle is 60°.

Practice

1. Draw a straight line, mark a point A on it, and construct an angle of 60° at A using only your compass and ruler. After you have finished, check the two arcs you drew had the same radius and that you joined A to the point where the arcs crossed.
2. Explain, in your own words, why the construction you just did gives an angle of 60° rather than, say, 45° or 90°. What must stay true about your compass radius for this to work?

Check your work

Attempt every question before you read this page.

Constructing Parallel Lines with a Set Square and Ruler

1. You placed the set square with one edge exactly on AB, held the ruler against the side of the set square, slid the set square along the ruler until its edge reached P, and drew the line through P. The line you drew is parallel to AB.
2. The two gaps you measured should be equal. This is exactly the test for parallel lines: they stay the same distance apart everywhere, so they can never meet.
3. All three lines are parallel to one another. The line through P and the line through Q are each parallel to AB, so all three run at the same angle and never meet one another.

Drawing Perpendicular Lines

1. Your constructed line meets PQ at a right angle. The point where the two lines cross is the point you chose on PQ. Check that the two lines form a clean 90° corner.

   Lay the ruler's edge along PQ. Press the set square's short edge against the ruler, slide it to your chosen point, hold it firm, and draw along the other short edge. The drawn edge is perpendicular to PQ because it follows the set square's right-angle corner.
2. All four angles where the perpendicular crosses AB are equal and each is 90°.

Measure AB, mark its midpoint, then place the ruler along AB and slide the set square to the midpoint. Draw along the set square's other short edge. Each of the four angles where the two lines cross is a right angle, 90°, because the two lines are perpendicular.

Cutting a Line Exactly in Half

1. CM = 4 cm and MD = 4 cm. The perpendicular bisector cuts the 8 cm segment exactly in half.
   a. Draw CD = 8 cm. 2. Put the compass point on C, open the compass more than 4 cm (more than half), and draw an arc above and below the segment. 3. Without changing the opening, put the point on D and draw another arc, so the arcs cross above and below. 4. Draw a straight line through the two crossing points. It meets CD at its midpoint M. Half of 8 cm is 4 cm, so CM = 4 cm and MD = 4 cm.
2. If the opening is less than half, the two arcs drawn from the two ends cannot reach each other, so they do not cross and there is no point to join. If it is exactly half, the arcs only touch at one point on the segment, so you cannot draw the perpendicular bisector through it. Opening it more than half makes the arcs cross in two places, one above and one below, giving the two points needed to draw the bisector.
3. It passes through the exact midpoint M of the segment, and it meets the segment at a right angle (90°). Because of that it is called the perpendicular bisector.

Constructing a Right Angle (90°)

1. Your construction should show a perpendicular line standing on the original line at O, and the angle between them should measure 90° with a protractor.

   Follow the worked example: draw the line and point O; draw a circle centred on O to cut the line at A and B; keeping the same radius, draw an arc from A and a crossing arc from B above the line, meeting at C; draw OC. The line OC is perpendicular to the original line, so the angle at O is 90°.
2. Keeping the radius the same makes OA = OB = AC = BC, so the figure formed is a rhombus whose diagonal OC splits the straight angle of 180° in half, giving 90° on each side.

Drawing an Angle of 60°

1. A correct construction shows an angle at A of 60°. Your two compass arcs must have the same radius, and the line from A should run to the point where those two arcs cross. If it does, the angle is exactly 60°.

   Draw line AB, then swing an arc from A across the line (meeting it at B). Keeping the radius unchanged, swing a second arc from B so it crosses the first (at C). Join A to C. Since AB = BC = AC (all one compass radius), triangle ABC is equilateral, so angle BAC = 60°.
2. The three sides AB, BC and AC are all equal because each is made with the same compass radius. A triangle with three equal sides is equilateral, and every angle of an equilateral triangle is 60°. The radius must stay the same (unchanged) throughout the construction, or the sides will not be equal and the angle will not be 60°.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-4257f18d	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____